

☒ HARD

☐ ALGEBRA

Algebra

6 questions · ~9 min

14

Q1

ALGEBRA

During a month, Morgan ran r miles at 5 miles per hour and biked b miles at 10 miles per hour. She ran and biked a total of 200 miles that month, and she biked for twice as many hours as she ran. What is the total number of miles that Morgan biked during the month?

- A. 80
- B. 100
- C. 120
- D. 160

$$4x - 9y = 9y + 5$$

$$hy = 2 + 4x$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h ?

- A. -9
- B. 0
- C. 9
- D. 18

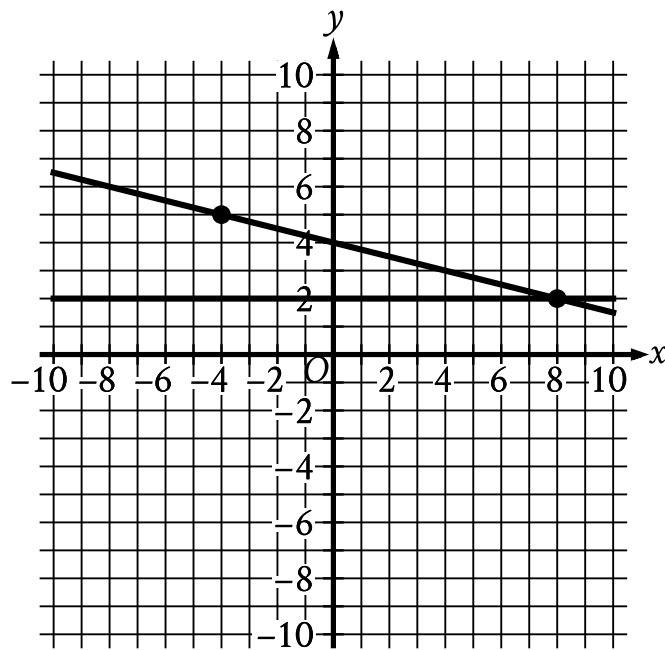
A sample of a certain alloy has a total mass of 50.0 grams and is 50.0% silicon by mass. The sample was created by combining two pieces of different alloys. The first piece was 30.0% silicon by mass and the second piece was 80.0% silicon by mass. What was the mass, in grams, of the silicon in the second piece?

- A. 9.0
- B. 16.0
- C. 20.0
- D. 30.0

$$\begin{aligned}8x + 7y &= 9 \\24x + 21y &= 27\end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

- A. $r, -\frac{8r}{7} + \frac{9}{7}$
- B. $-\frac{8r}{7} + \frac{9}{7}, r$
- C. $-\frac{8r}{7} + 9, \frac{8r}{7} + 27$
- D. $\frac{r}{3} + 9, -\frac{r}{3} + 27$



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (negative 4 comma 5)
 - (0 comma 4)
 - (8 comma 2)
- For the second line in the system:
 - The line is horizontal.
 - The line passes through the following points:
 - (0 comma 2)
 - (8 comma 2)

If a new graph of three linear equations is created using the system of equations shown and the equation $x + 4y = -16$, how many solutions x, y will the resulting system of three equations have?

A. Zero

- B. Exactly one
- C. Exactly two
- D. Infinitely many

Q6

GRID-IN ALGEBRA

$$x - 2 - 4y + 7 = 117$$

$$x - 2 + 4y + 7 = 442$$

The solution to the given system of equations is x, y . What is the value of $6x - 2$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.